

Simple Insertion Sort: A Simple insertion sort is one of the insertion sort techniques, which sorts a set of records, by inserting records into an existing sorted file.

The C function for Simple Insertion sort is given below:

```
void insertion_sort(int a[], int n)
{
    int i,j,temp;
    for(i=1;i<n;i++)
    {
        temp=a[i];
        for(j=i-1;j>=0 && temp<a[j];j--)
        {
            a[j+1]=a[j];
        }
        a[j+1]=temp;
    }
}
```

Input Array:

45, 33, 55, 28, 66, 18

I Pass	33, 45, 55, 28, 66, 18
II Pass	33, 45, 55, 28, 66, 18
III Pass	28, 33, 45, 55, 66, 18
IV Pass	28, 33, 45, 55, 66, 18
V Pass	18, 28, 33, 45, 55, 66

Efficiency of Simple Insertion Sort:

If the initial numbers are sorted, only one comparison is made, on each pass, so that the sort is $O(n)$.

If the initial numbers are sorted in reverse order, the sort is $O(n^2)$, since the number of comparison is $(n-1) + (n-2) + \dots + 3 + 2 + 1 = n(n-1)/2$